Result 1 (graded induced component). [Topology foundations, graded induced map component]
Let K be a type. Let *field* be a field structure on K. Let X and Y be topological spaces. Let g be a continuous map from X to Y. Let p be a natural number. Then

$$H_*(g; K).\text{component}(p) = g_*.$$

Result 2 (graded induced identity). [Topology foundations, singular homology functor identity]
Let K be a type. Let *field* be a field structure on K. Let X be a topological space. Then

$$H_*(\text{id}_X; K) = \text{id}_{H_*(X;K)}\,.$$

Adapted from the cited source.

Result 3 (graded induced composition). [Topology foundations, singular homology functor composition]
Let K be a type. Let *field* be a field structure on K. Let X, Y and Z be topological spaces. Let f be a continuous map from X to Y. Let g be a continuous map from Y to Z. Then

$$H_*((g \circ f)\,; K) = (H_*(g; K) \circ H_*(f; K)).$$

Adapted from the cited source.

LLM-assisted example and proof (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.